

A Visualization Tour of Start-Up Financing, Investments, and Takeovers: Venture Capital Trends

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Abstract - The project is based on the visualization of the global startups financing and industry trends based on the 1900-2014 years data we retrieved from Crunchbase in the Kaggle. We made interactive Tableau visualizations with the several cleaned and merged excel files to study the four main domains, i.e., industry growth, funding stages, geographic hotspots, and a built-in dashboard. The findings reveal that Information Technology, Software, and Biotech are main categories of the process of formation of startup and the overall fundings, and the United States is the most common center of startup activity with growing activity in Europe and South-East Asia. The project also offers precise, quantitative insights which can assist the entrepreneur, investor and researcher comprehend past trends in the innovation and venture capital.

I. Introduction

Startups over the world has experienced a drastic changes during the last century which was influenced by changing priorities of the industry, the speed of the technological revolution, and the emerging trends in terms of venture financing. It takes more than raw data to comprehend such changes it requires clear interactive graphs that show the flow of funding across industries, regions, and even growth phases. This project also satisfies the said need, by combining the datasets gathered by Crunchbase in a single system of Tableau dashboards, where the user will be able to search the industry trends, distribution of funding stages and geographic hotspots in a meaningful and intuitive manner.

A. Background and Relevance:

This project is motivated by the fact that the global startup ecosystem is becoming more and more complex, and venture capital becomes more significant in the context of technological innovation. Big investment moments like multi-billion-dollar AI funding rounds and massive acquisitions- all show the use of venture capital in boosting the expansion of newer industries. However, a large portion of this data is dispersed in a collection of data sets and cannot be easily analyzed without visual programs. This project will fill a genuine information vacuum: the requirement to find pertinent, empirical information on the long-term trends in the industry and changes in the geographical locations of startup activity. It can be applied to the entrepreneur who is going to get opportunities, the investor who is going to analyze the market and the researcher who is going to analyze how the system of innovation is evolving. The project will convert messy historical data in an easy-to-use resource format through interactive dashboards and visual analytics that will facilitate better cognition and decision-making in the venture capital sector.

B. Target Audience:

The main target audience of this visualization project is entrepreneurs, investors, analysts, researchers, and policymakers that need the clarity of the insights into the startup and venture capital ecosystem. Visualization can help entrepreneurs comprehend what industries receive the greatest investments and in what new ecosystems startups are being created so that they know where to grow. The ability to easily identify high levels of activity in the funding process, analyze market trends, and determine the areas which are gaining momentum is beneficial to investors and venture capital analysts. The dashboards can be used by researchers, students, and policymakers to learn about the prevalent trends of innovation diffusion, economic development, and long-term industry transitions in different parts of the world. The visualizations designed in Tableau are both user-friendly and intuitive, and the findings can be widely utilized by both technical and non-technical users and can be readily interpreted.

C. Motivation and Significance:

The accelerated technological changes, acquisitions in the billions of dollars, and the changes in the priorities of the global investments define audiences of modern startup ecosystems. The large-scale venture capital events, e.g., multibillion-dollar investments in AI, acquisitions in design and tech, represent how venture capital provides a jumpstart to whole industries. Nevertheless, the majority of the information concerning funding round, industries, and geographic distribution is still scattered across various sources, and affiliated stakeholders will not be able to receive the whole picture. The proposed project fills this gap by structuring, scraping, and incorporating intricate datasets based on Crunchbase into an interactive visualization space. The importance is the revelation:

- What industries have long been the most funded ones?
- The trend of funding stage (Seed, Series A/B/C).
- What regions are becoming international sources of innovation.

These lessons will be useful in making better decisions and enhance the knowledge of how venture capital dynamics influence technological and economic development

D. Data set Description:

The project makes use of the startup investment data provided by Crunchbase acquired on Kaggle, which is stored as several CSVs, including objects.csv, fundings.csv, offices.csv, investments.csv, acquisitions.csv, and ipos.csv. These data sets are 1900-2014 and were cleaned, merged, and normalized into one analytical data set. The most significant preprocessing steps were currencies to USD conversion, standardization of dates to the year of founding, grouping fund rounds into major phases (Seed, Angel, Series A-C), and managing the missing data as well as adding geographic coordinates in offices.csv. This aggregated data is the basis of any industry, funding stage and geographic visualization in our project. The preprocessed Final dataset yielded to be 22891 entries and 31 features. Some of the null values we are not able to handle or remove and hence we are filtering them when using the dataset in tableau.

	id	entity_id	normalized_name	major_industry	category_code	permalink	founding_year	country_code	state_code	city	...	funding_partial	funding_c	funding_unattributed
0	c:1	1	wetpaint	Web & Internet Services	web	/company/wetpaint	2005	USA	WA	Seattle	...	0.0	25000000.0	0.0
4	c:1001	1001	friendfeed	Web & Internet Services	web	/company/friendfeed	2007	USA	CA	Mountain View	...	0.0	0.0	0.0
6	c:10014	10014	mobclix	Other / Miscellaneous	mobile	/company/mobclix	2008	USA	CA	Palo Alto	...	0.0	0.0	0.0
7	c:10015	10015	fitbit	Health & Biotech	health	/company/fitbit	2007	USA	CA	San Francisco	...	0.0	12000000.0	45069200.0
8	c:100155	100155	mtpv	CleanTech & Energy	cleantech	/company/mtpv	2003	USA	TX	Austin	...	0.0	0.0	0.0
5 rows × 31 columns														
Shape of df_with_funding_rounds: (22891, 31)														

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E. Objectives:

- **Observing Industry Trends:** A scan of long-term formations of startups, levels of funds and industry dominance.
- **Visualize Funding history across Series:** Decomposition of the venture capital activity by Seed, Angel, Series A, Series B and Series C: frequency and size of the capital raised.
- **Mapping Geographic Hotspots:** Also find world and country centers of startup action and mark those areas that are getting a heavy dose of venture funding.
- **Develop an Interactive Dashboard:** Visualizations are brought together into a single Tableau dashboard, which has filters based on industry, geography, funding stage and time.

- Naga Sampath Maddineni:** I have collected the dataset files from Kaggle. This has multiple files like objects.csv, fundings.csv, offices.csv, investments.csv, etc., from the timeline of 1900 to 2014. I have not got the recent data, as it needs a subscription. We have 11 such CSV files in the data. Using the combined dataset which has all the necessary columns given by my teammate Abhiram . Next, we have Normalized the columns like the date to year we mainly require for our analysis, normalized the funding stages keeping

only the main ones like seed, series a, series b, series c, angel. Changed the currencies to USD nominal. Next, I have done primary visualizations in tableau to see the industry trends using the using Geographical hotspots and built website for Objective 2,3 and interactive Dashboard 2

- **Abhiram Pudi:** We have combined the data files collected by my teammate Sampath using company id as key across all the files. Removed the product details as we focused only on the companies from the dataset. Handled the missing values, divided the fundings series added them to the dataset. Merged the offices.csv getting the locations, country codes, state codes, region, city, latitude and longitude which will be helpful in analyzing the Geographical trends which is one of the objectives. I have also conducted the EDA analysis on the Total fundings, category codes, to see which categories are more in the startups. Next, I helped my teammate Surya to see and reference the papers to get more ideas. Furthermore, I helped with building the website for Objective 1, and interactive Dashboard 1
- **Bandaru Sai Sri Surya:** Helped my teammates Sampath and Abhiram with the dataset creation, by cross checking the fields, analyzing and giving the columns that are needed to fulfill our project objectives. We performed the statistics in collab itself to see and check the numerical column analysis like total fundings, and we created a separate dataset named startups funded excel file which all the funded startups which is helpful for my task. And worked on the acquisitions to see yearly acquisitions counts using tableau. Worked on Plots in the tableau for the series funding analysis of objective 2 and Helped in the creating the interactive dashboard 2 along with my teammate Sampath.

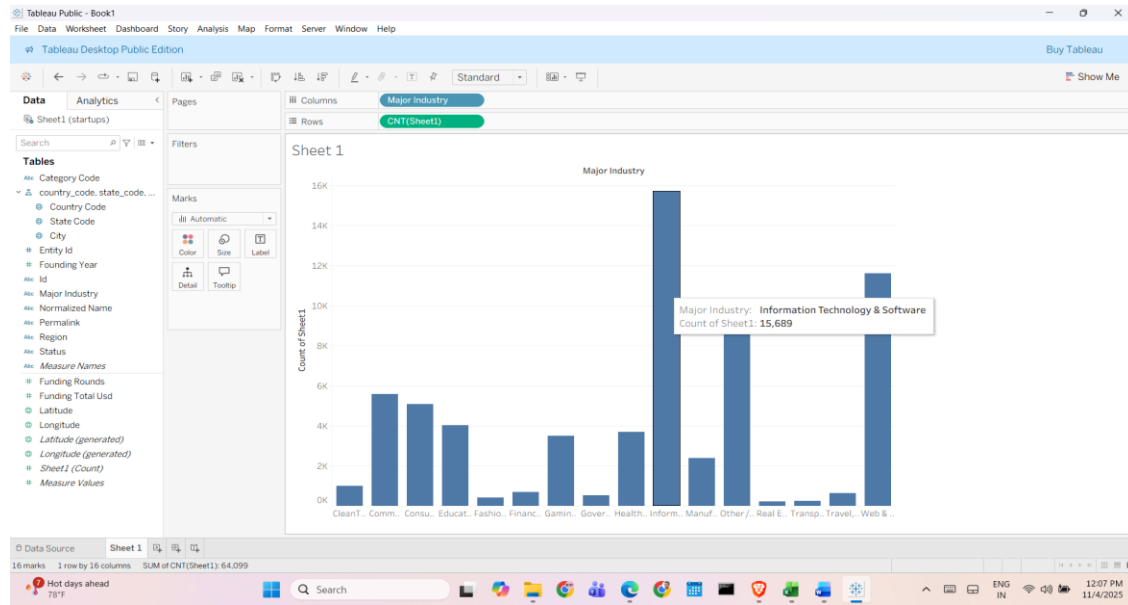
II. Visualizations Functions:

A. Objective1:Visualization of Industry Trends and Explanations

This objective is mainly about finding out the evolution of different industries in the startup companies over a time period. By comparing the no. of startups, the total fundings, and the number of investments per year, we will know what sectors were prevalent in the past and the way investments changed over the decades and their priorities. Those visuals indicate the long-term trends, e.g., the emergence of Information Technology, Software, and Biotech, and allow people to see how the trends of innovation and investor interest have fluctuated in various industries since 1900 up to 2014.

1. Bar Chart of Distribution of Startups across Major Industries:

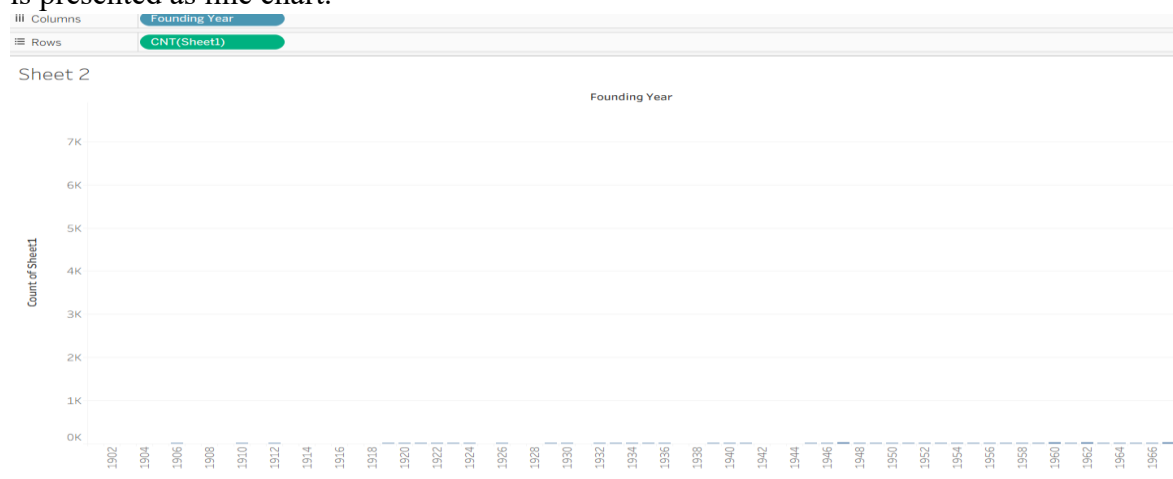
In this plot we are visualizing the counts across the major industries; we can see which industries are more prevalent.

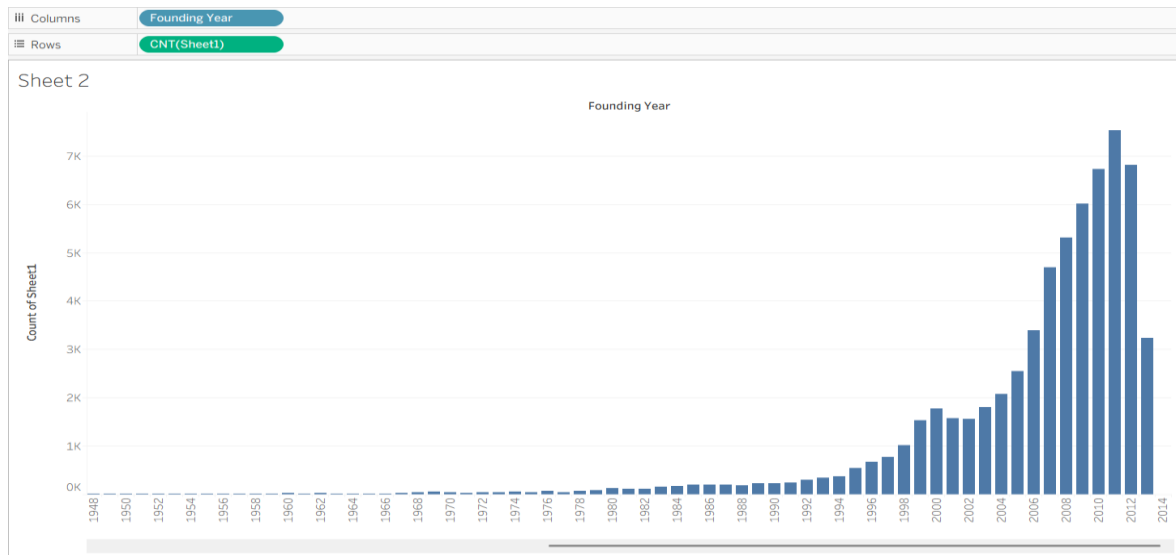


In the last Century, We can see IT and Software industry has the highest number of the startups, and this implies that it was the leading industry during the period, followed by the Biotechnology industry. This shows how the use of computers and other digital technologies in the time around the beginning of the 21st century shaped the startup environment, becoming the heart of innovation and the source of new opportunities in various spheres. However, on the other side we can see that industries like the Real Estate and the Transportation industries had the least number of startups in this period, which shows that there is less entrepreneurship in these industries compared to the technology-based sectors.

2. Vertical Bar Chart of No. of Startups Started across the Years:

This chart will give us the idea of which years has seen more startups appearing, In website it is presented as line chart.

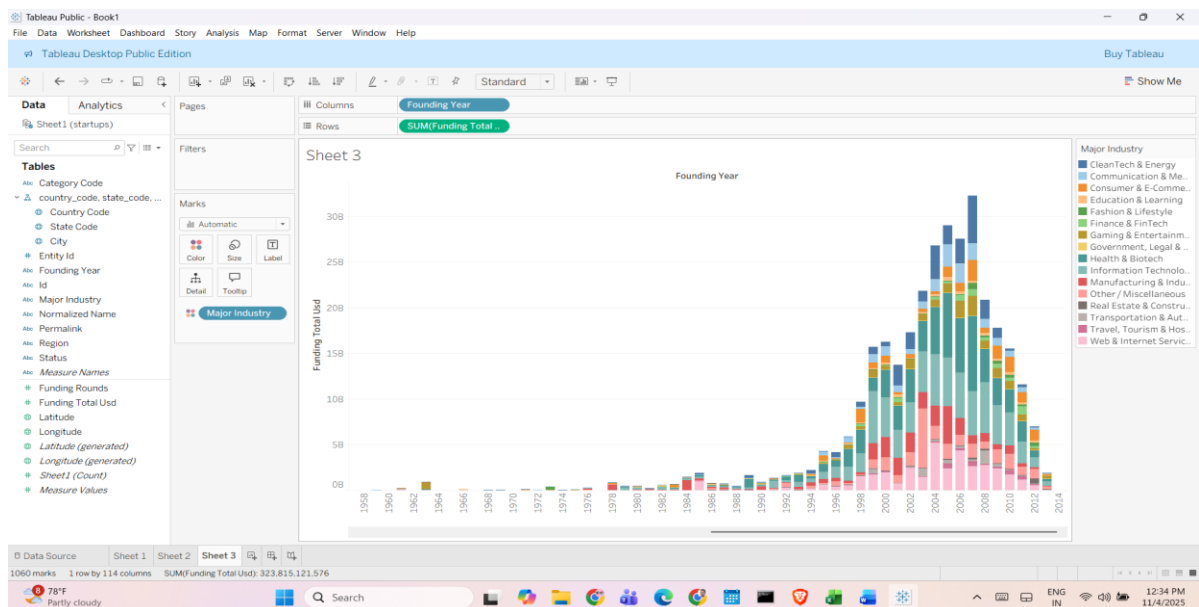




As we can observe that the years between 1902 and 1970 are nearly dry with no startups in the beginning of the century, the concept and industries have begun to emerge in the early 1970s that resulted in the startup revolution in the industry sectors. As we can observe in the picture below, we can say that there is gradual rise of the startups between late 1900s-early 2000s. So does the startup industries and the years they were started during the century.

3. Stacked Bar Chart For founded Year vs Fundings with Major industries in color:

This gives the idea of fundings across different years for different industries.



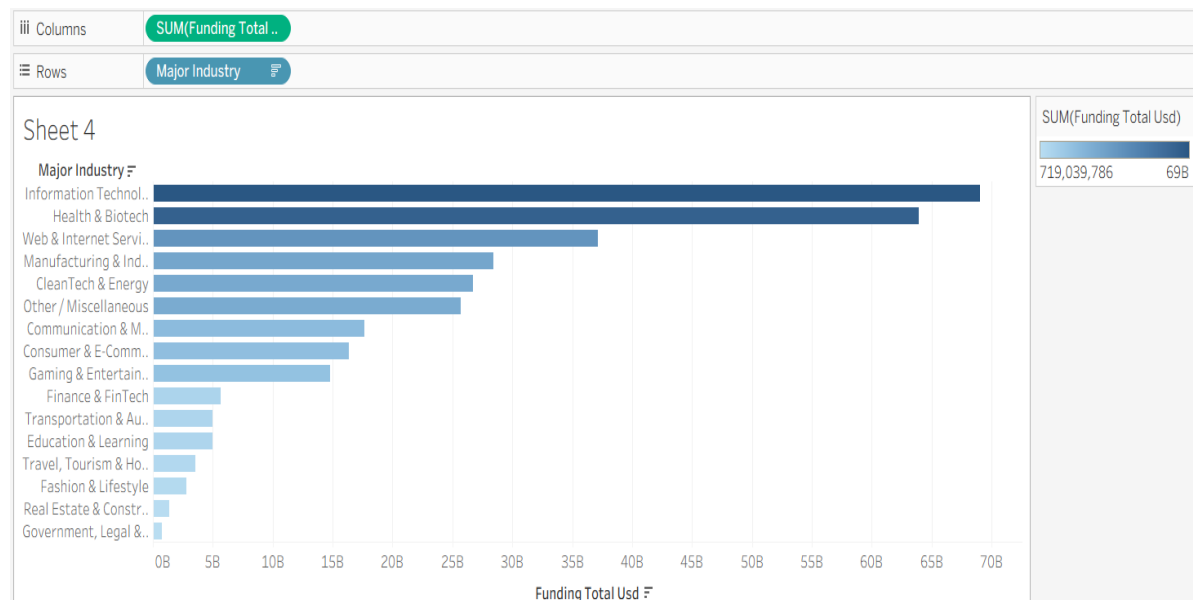
As can be observed, the amount of startup capital is not evenly distributed throughout the years, but rather, there are steep spikes of this amount that represent the eruptions of the high investment activity. The earlier years of founding, 1910s to 1950s, have sporadic-but-high value investments, especially within the segments of Consumer and E-Commerce, Cleantech

and Energy and Information Technology and Software. These spikes are signs of the appearance of industrialization and the beginning of the technological ventures, which gained large amounts of capital even when the number of startups was smaller.

Towards the end of the 20th century and early 2000s, funding patterns become more diverse, with such emerging industries as Health and biotech, Fintech, and Web and Internet services becoming prominent. The color difference in the later years shows the change of priorities in ventures to knowledge-based and digital-focused sectors, signifying the change of the conventional manufacturing and energy start-ups to the technology-oriented innovation centers. The cumulative investment amounts to more than one billion USD in the major years such as 1953 and 2003 depict clumped mega-rounds or large takeovers in these industries. In general, visualization proves that investment priorities have changed throughout the years, and technology, health, and consumer-oriented industries are the most popular in the modern startup ecosystem.

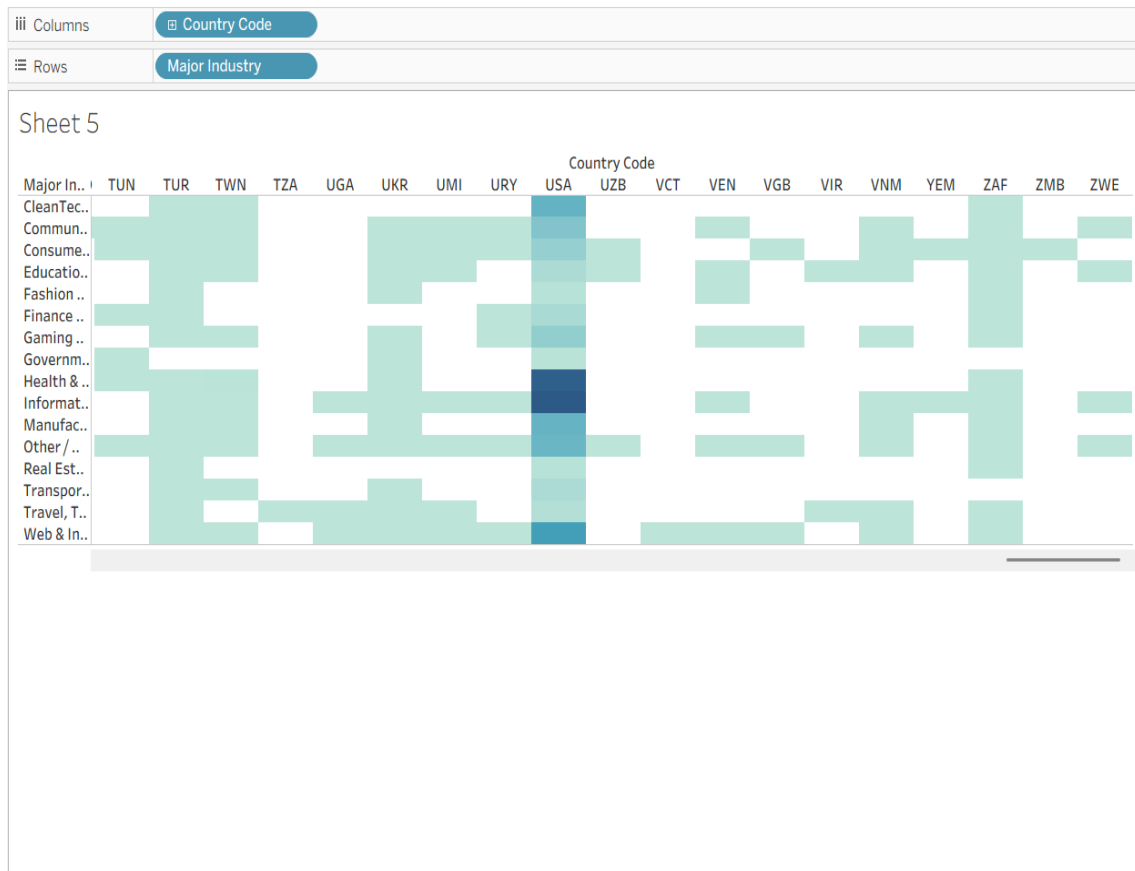
4. Horizontal Barchart of funding VS Major Industries

This gives the idea about the top industries by total funding acquired. As you can see in chart below:



According to the bar chart, Information Technology and Software has the highest total funding of approximately \$70 billion USD which is closely followed by Health and Biotech which has a total funding of about 68 billion USD. Web & Internet Services takes the third position of about \$40 billion USD and Manufacturing and Industrial Tech and Cleantech and Energy have attracted between 25-30 billion USD and 25-30 billion USD respectively. Conversely, other industries such as Real Estate, Fashion and Lifestyle and Government and Legal have a below 5 billion USD which implies that majority of venture capital is still concentrated in technology induced industries.

5. Heatmap of Countries Vs Industries with Funding Total to display intensity:



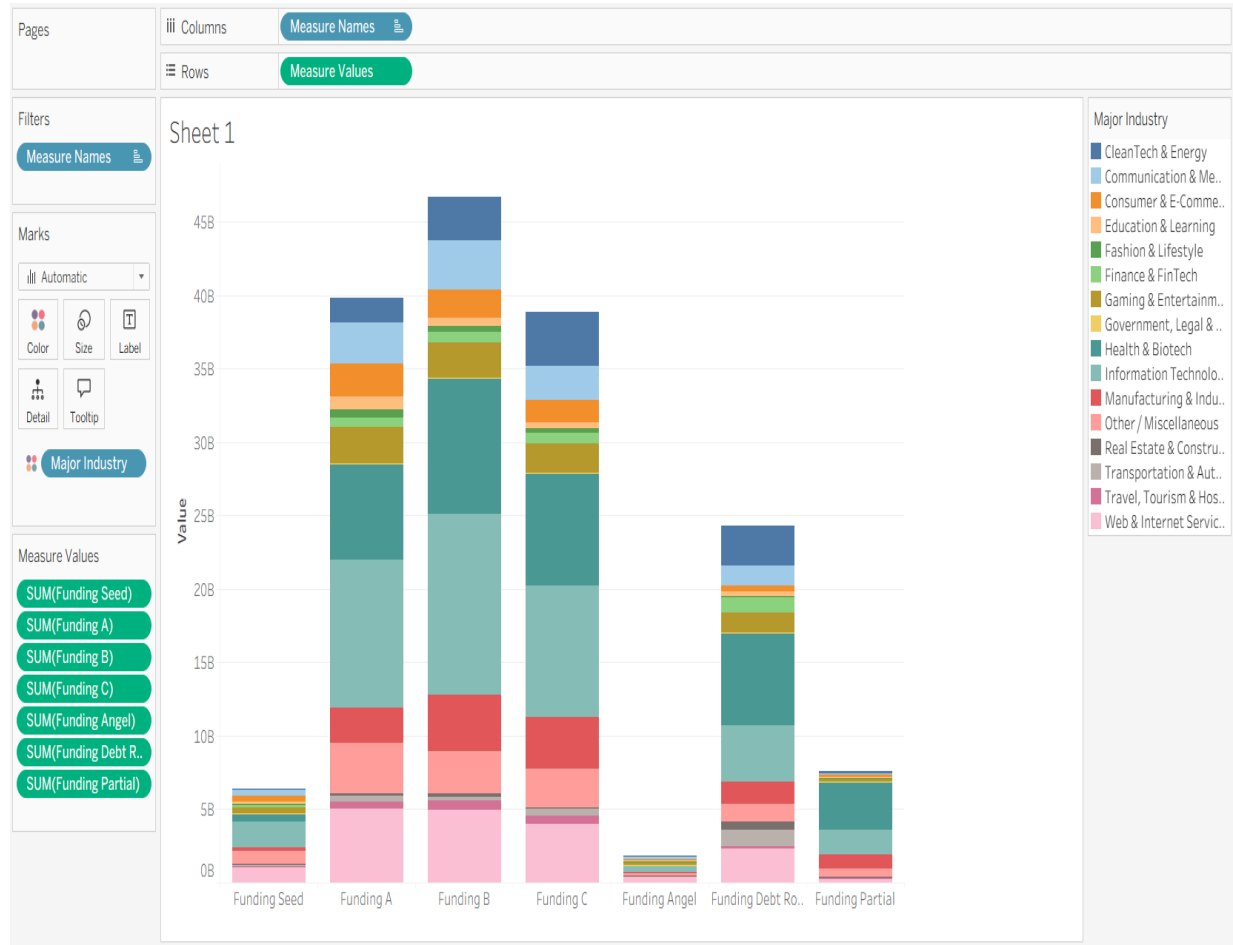
The heat map gives a picture of the distribution of venture funding in industries and countries. The darkest blue shades indicate the highest level of concentration of investments in the USA, especially in the spheres of Information Technology and Software, Health and Biotech and Web and Internet Services. The activity of moderate funding is also observed in such regions as South Africa (ZAF) and Vietnam (VNM) in Cleantech and Manufacturing segments. All the other countries appear to be shaded less, indicating the presence of smaller or developing start-up ecosystems. In general, the evidence confirms that the venture capital is still geographically concentrated, and the United States is the focal point of the innovation-driven industries and only partial yet increasing involvement of developing economies.

B. Objective 2: Visualizations of the Fundings Across Various Series:

This one is aimed at getting insights into the process of the movement of capital through the various phases of the startup lifecycle, including early stages of the lifecycle, such as Seed and Angel, and growth stages, such as Series A, B, C. Analyzing the number of startups at each stage as well as the total amount of startups financed we point at the stage where the attention of investors is concentrated and the flow of startups financed.

1. Stacked Bar Chart of Series Funding Amount VS Stages and Industry in the colors

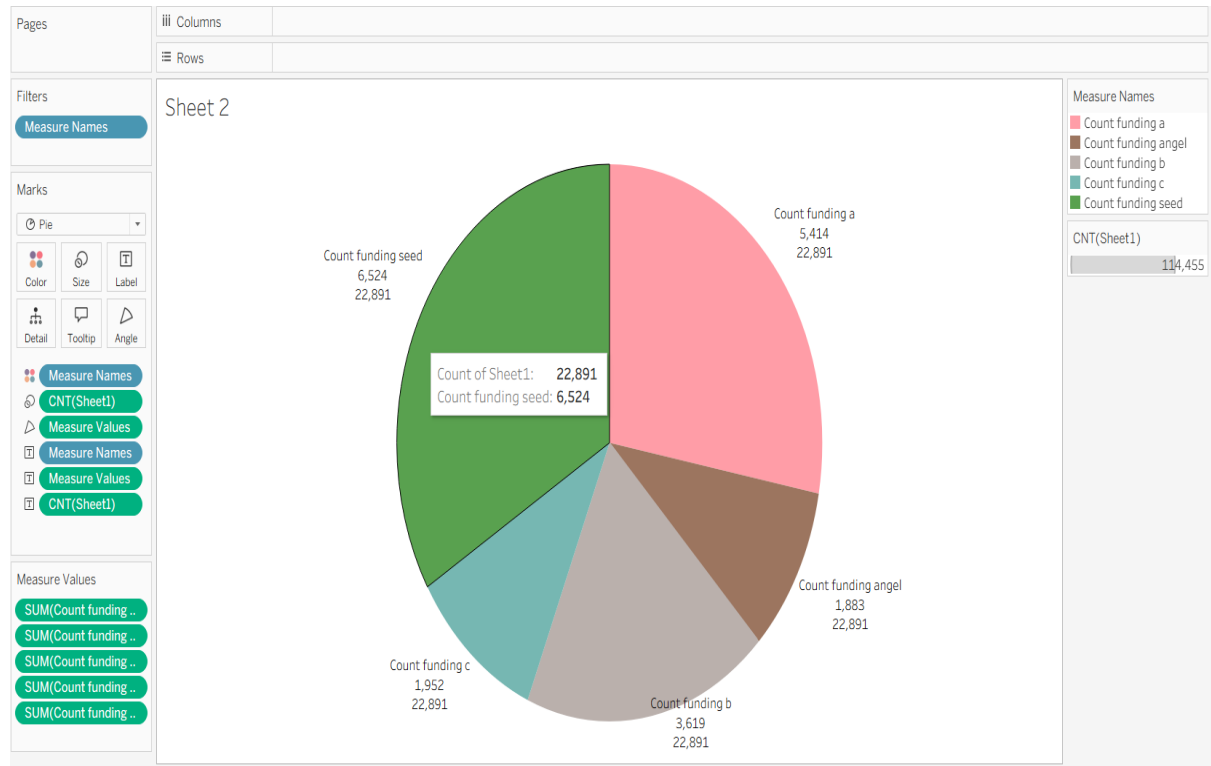
Breaks down all funding by major stages (Seed-Series C) and illustrates industries receiving the most funding at each stage.



As the above stacked bar chart demonstrates the allocation of the startup funding between various stages and industry. We can say that the biggest sources of funds are the Series B and Series A rounds, which are over 40-billion USD with each source, which shows that investors have been active in growth-stage startups. This is closely followed by series C, which is not new in terms of scaling ventures with about 35 billion USD. Earlier rounds like Seed and Angel have relatively smaller investments of less than 15 billion USD, but they cover a broader industry. Also, the Debt and Partial rounds adds a little bit to the overall financing scene. The top performing industries, by most levels, include Information Technology & Software, Health and Biotech, and Web and Internet Services and this highlights the venture capital preference of the digital scalable and innovation-focused industries. Generally, we conclude that the visualization confirms that middle-stage rounds are the areas where the intensity of funding is the highest, and the early and late rounds are in supportive positions within the lifecycle of the startups.

2. Pie Chart of the distributions of Startups for Funding Series(Seed to Debt)

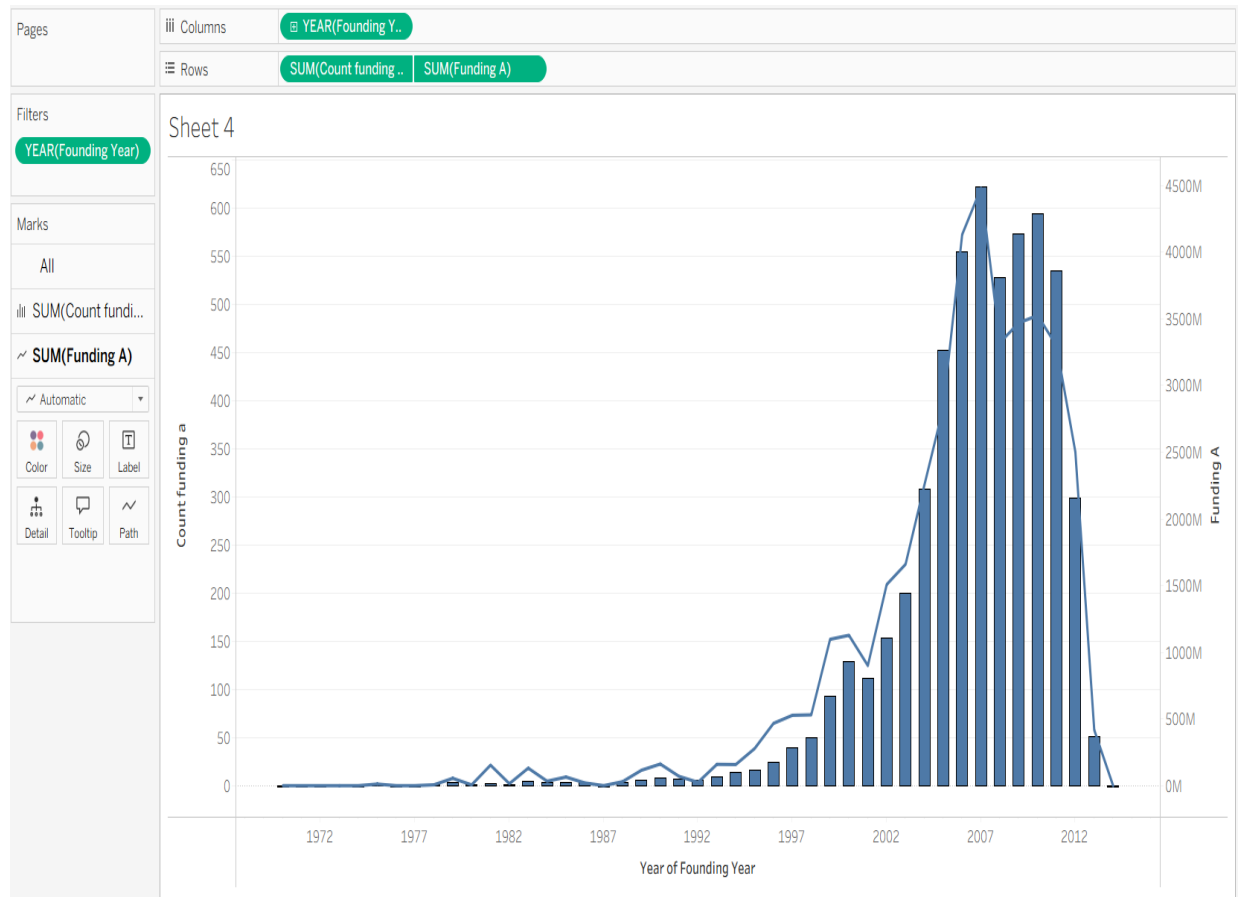
presents the number of startups that reach each financing round with the majority of companies being at an early stage including Seed and Series A.



The pie chart shows how the startups are distributed according the level of funding that they are at. The largest proportion of all the companies observed is the Seed-stage start-ups taking up about 28-30% of the total number of firms, after which there is the Series A taking up approximately 24-26%. It has a contribution of about 16% by Series B, 10-12% by Series C and Angel-funded startups respectively. It means that the majority of startups only obtain early-stage investments, which are active and easily accessible in terms of Seed and Series A funding rounds. Yet, the ratio of startups declines dramatically at later stages, i.e. Series B and C, which underscores the selective nature of the advanced venture financing and the pipeline narrows as startups become older.

3. Combination Chart: Series A Funding Trends Over Time:

Shows the number of startups that have received Series A financing as well as the cumulative capital that is raised annually.



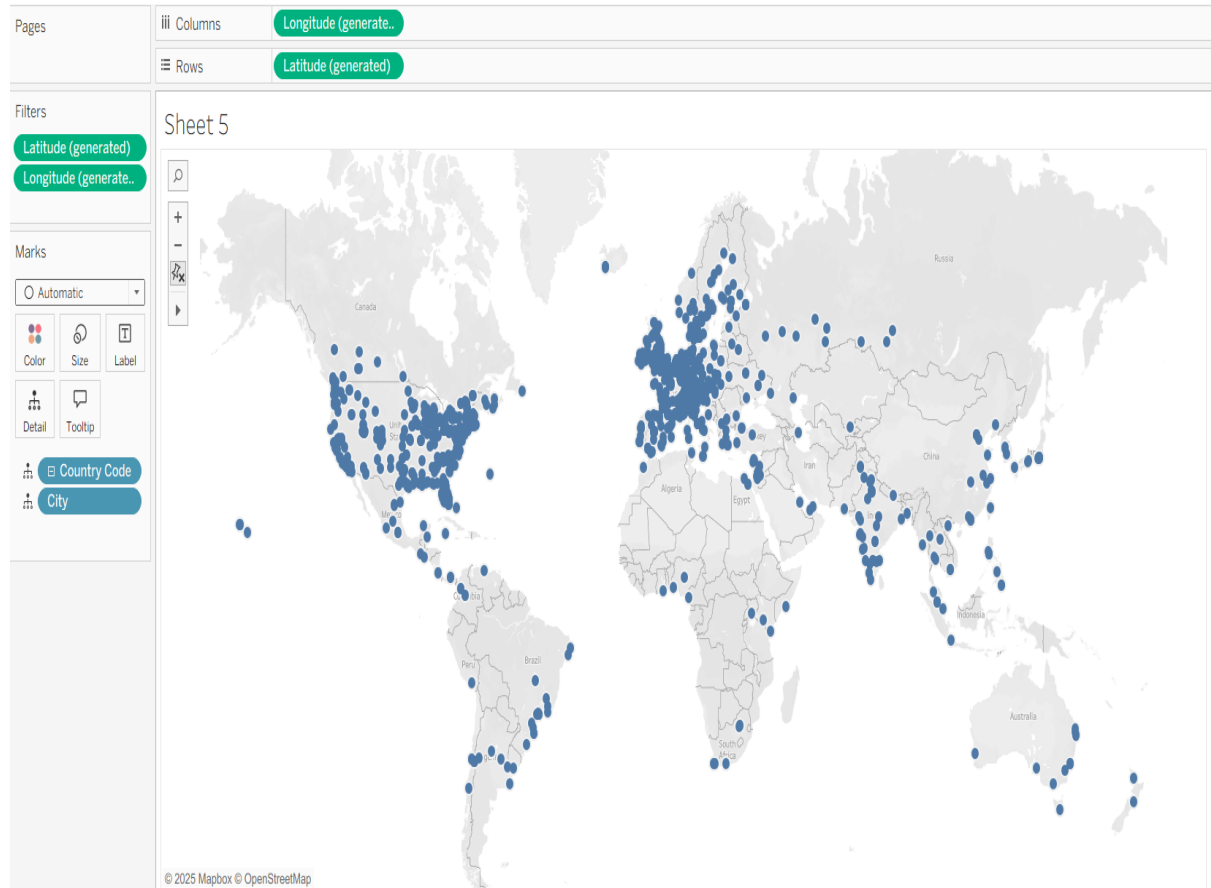
The combination chart shows a history of Series A funding activity between 1970 and 2015 with bars on the left axis demonstrating the number of startups funded and line on the right axis showing the total volume of Series A investment by the year of founding. This information shows that until the early 90s, Series A funding was small, but has steadily increased since that time, approximately in 1997 when the internet and technical startups began to grow. The highest activity was reached between 2005 and 2010 when the number of companies as well as the volume of investment were at the highest level of record-breaking 4,000 million USD in capital. There is a noticeable drop in both measures after 2012, which can be attributed to a lack of data on newer firms or a post-boom fall in the start-up investment. In general, visualization brings out the fact that the Series A investments increased in the mid-2000s during the technology boom, which was one of the major growth stages in the history of venture capital.

C. Objective 3: Geographical Hotspots

This goal will be achieved by revealing where the activity of startups is the most concentrated in the world. Geographic visualizations allow us to determine global and national centers of innovations, emphasize the regions where entrepreneurial ecosystems are growing, and compare the level of funding in each country and city to understand how venture capital is incorporated geographically.

1. World Startup Location Map (Global Scatter Plot):

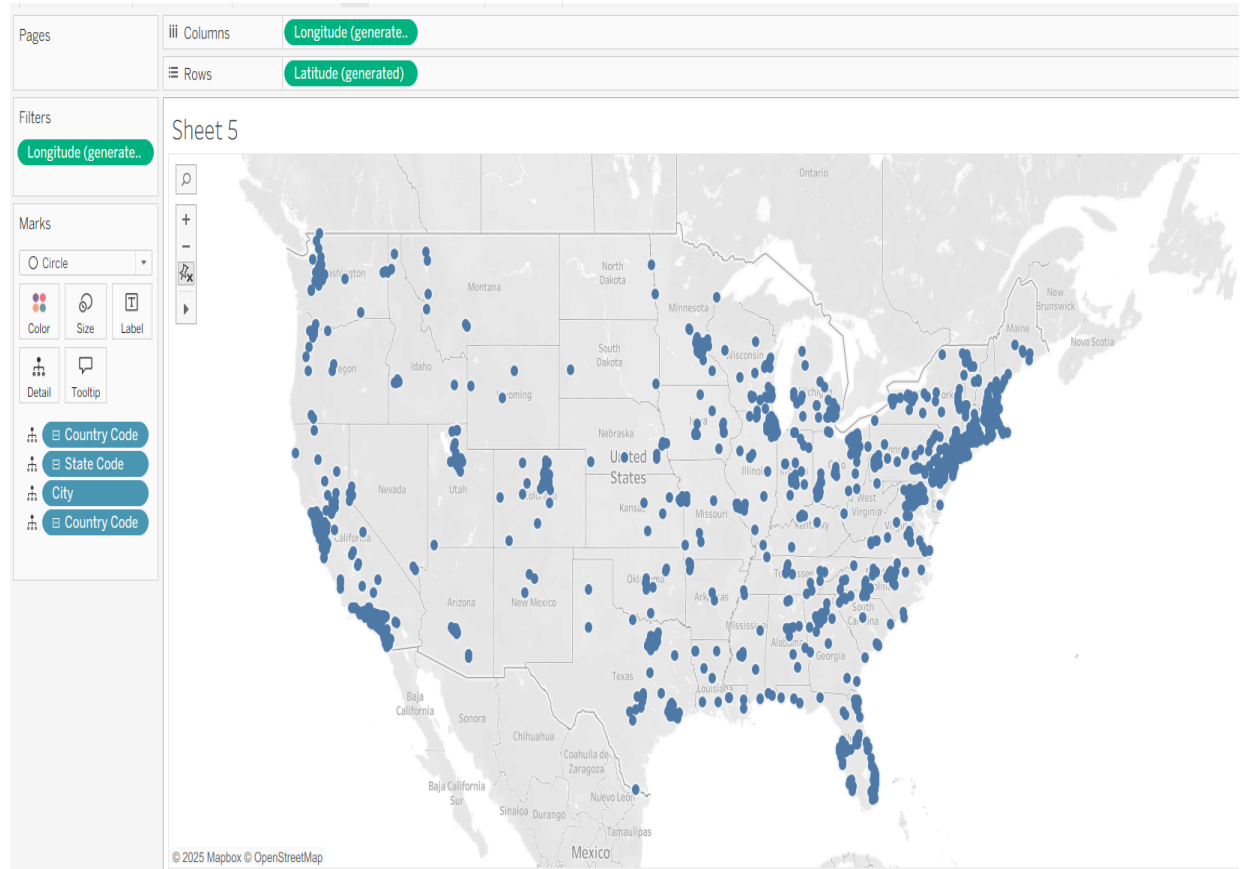
Determines global ecosystems of startup activity, which appear highly concentrated.



The above map provides a geographical distribution of the start-up locations around the world, which indicates evident geographic clusters, in which the entrepreneurship activity is high. We can see that the North America and Europe are shown as the most populated areas, and specifically the area around the United States- more so the East Coast, the West Coast, and while the Midwest areas in particular shows a high concentration of startup population. Europe presents equally thick clusters at Western and central parts, with the key areas of innovation that include the UK, Germany, France, and the Nordics. We can also see Asia has isolated but prominent points especially in India, China and Southeast Asia, which is an indication of a new startup ecosystems. South America, Africa and Australia have less but still significant clusters indicating less but increasing involvement in international venture activity. Comprehensively, visualization shows how the distribution of startup ecosystems all over the world is uneven, with high levels of concentration in the developed economies and growing presence in the developing areas.

2. U.S. Startup Density Map:

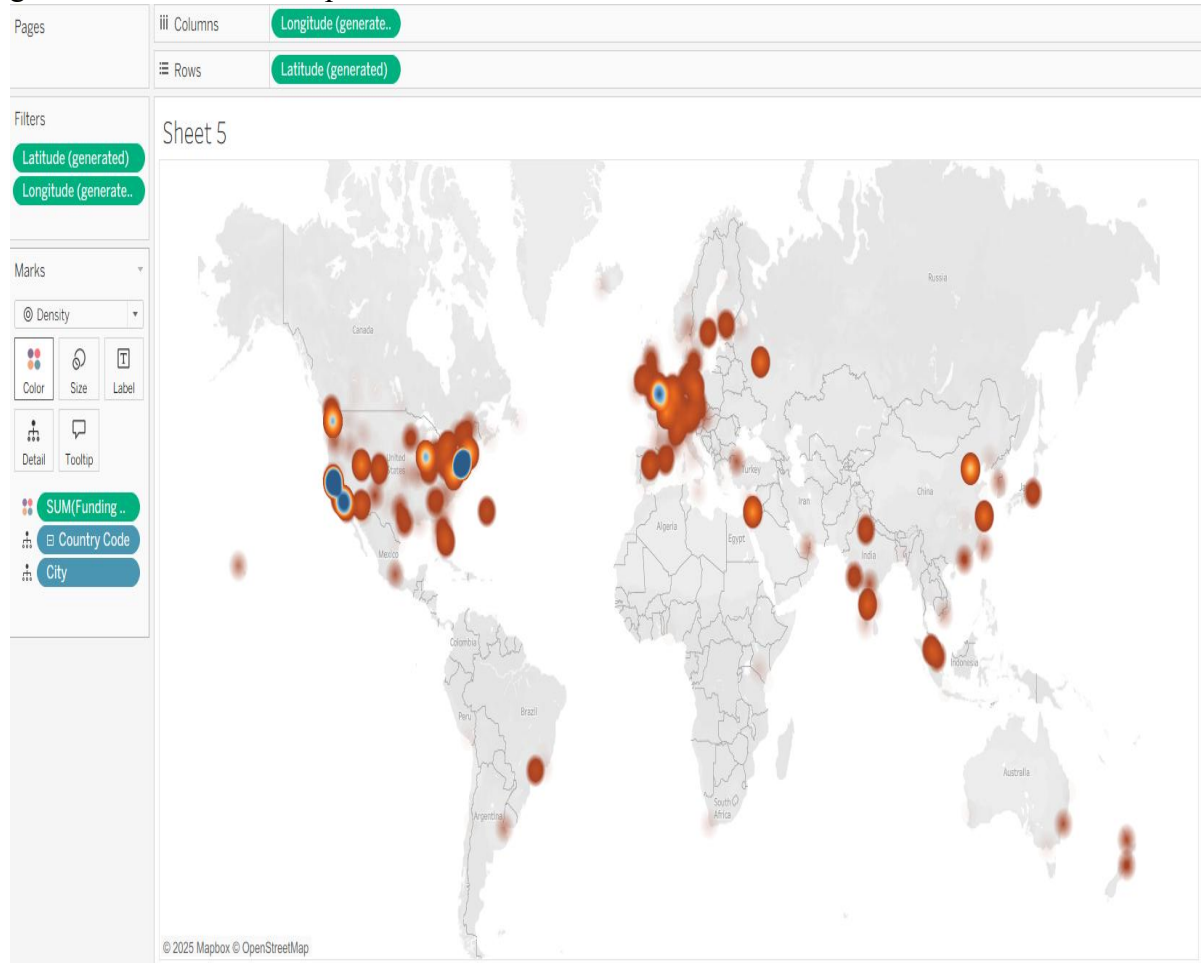
Highlights key U.S. innovation hubs and give idea of which states are good no of startups and friendly.



This map gives a pinpoint location of the spatial distribution of startups throughout the United States, and indicates the regions in which most of the entrepreneurship activity is located. The highest concentrations of clusters run along the East Coast - namely New York, Boston, and Washington, D.C. signifying a robust innovation hub with a large number of startups. Other prominent activity is in the West Coast, especially in California in the Silicon Valley, Los Angeles and Seattle which are renowned technology centres. States such as Texas, Colorado, Georgia and Illinois have impressive, but more diffusely distributed clusters of startups in the center and south indicating the development of ecosystems beyond the traditional coastline hubs. The fact that so many points have been scattered out to almost every state demonstrates that the activity of startups is not confined to the major cities but is gradually beginning to be spread into the mid-sized and emerging metropolitan areas. On the whole, the map highlights the geographic diversity of the U.S. startup ecosystems and nevertheless captures the prevalence of the coastal innovation hub.

3. Global Funding Density Heat Map:

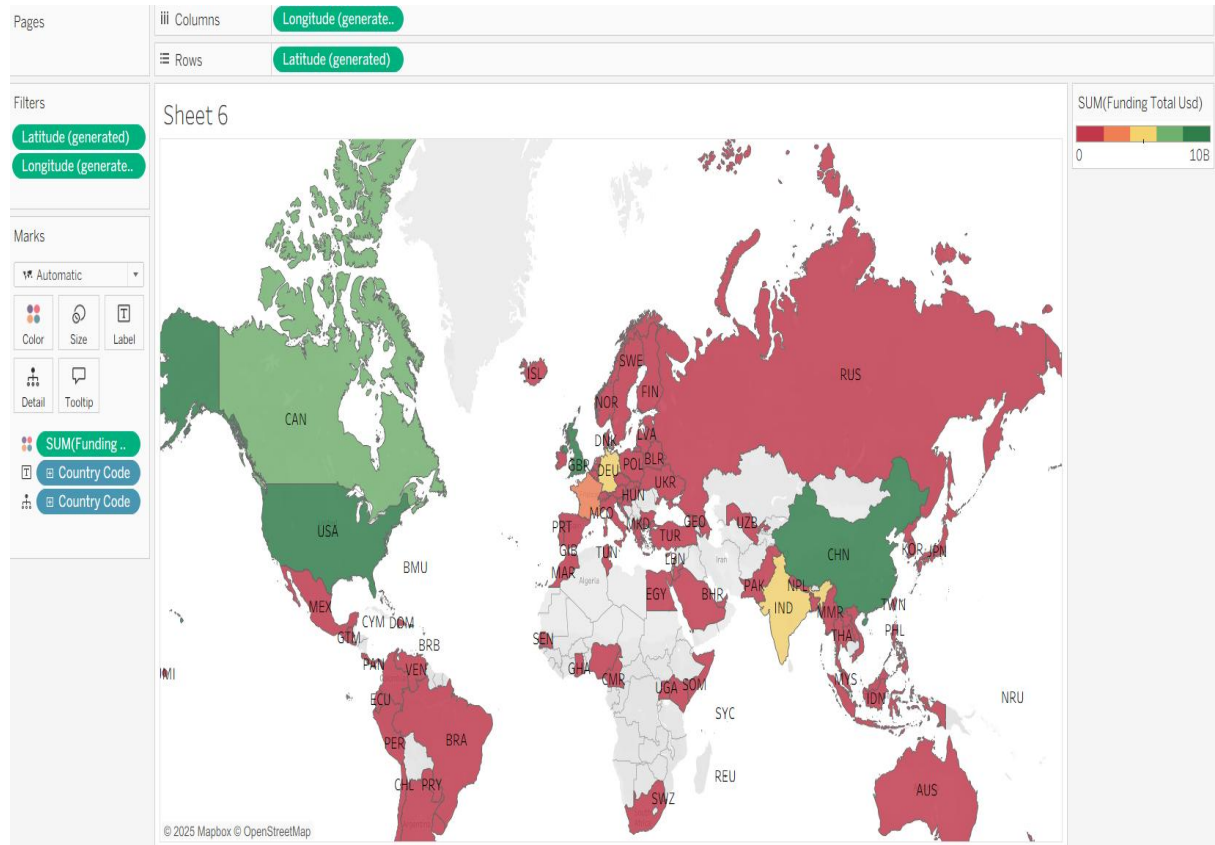
Exposes areas that have received the most in terms of cumulative startup capital showing key global foci of venture-capital concentration.



This density map brings to light the level of startup funding in various global regions and the hotspots are noticeable with the activity of investment being very high in these locations. The most noticeable clusters can be found in the United States, namely Silicon Valley, East Coast, major tech cities in Texas and Washington, which show where there are multiple startups as well as significant amounts of overall funding. Europe is the other region of high activity, especially around the UK, Germany, France and parts of Scandinavia, indicating established and well-developed innovation eco-systems. In Asia, China and India have strong concentrations of funds, where they can be seen to have fast growing markets with huge venture capital inflows. The density marks are normally smaller in size and lighter in color which makes identifying the areas with the most cumulative investment easier where the darker colors of orange reflect the larger amounts of funds invested. In general, the map visually highlights not only locations of startup but the highest financial activity, which can be used to identify the world hubs of innovations and capitalization.

4. Global Funding Choropleth Map:

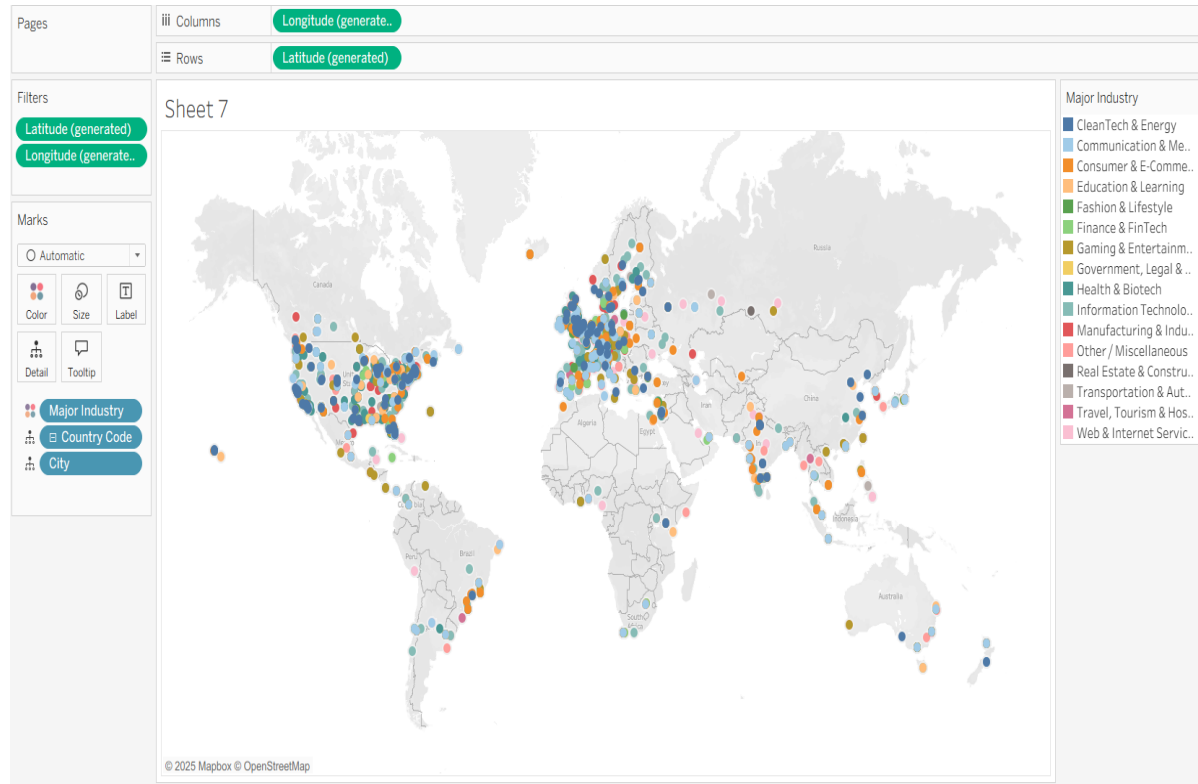
Demonstrates that country-specifics in total startup funding with high ecosystems in the U.S., Canada, China, and some European countries.



The map is a choropleth map which stylizes the total amount of startup funds allocated to individual countries with the intensity of colour used to emphasize those areas with the largest and the lowest amounts of investment. Countries such as the United States, Canada and China are presented in dark green and this means that these countries have exceptionally high venture capital activities and fully developed start up ecosystem. There are moderate levels of funding in India and some Western Europe which are indicated by yellow and light-green colors indicating high but relatively less robust investment environments. On the other hand, most nations in South America, Africa, Eastern Europe, and some parts of Asia are colored red, which means that little funds were used in starting up their businesses as reported in the data. This disparity is a vivid demonstration of access to venture capital in the world, where much financial resources are concentrated in North America, Western Europe, and key Asian economies and much parts of the world are either underfunded or are new entrants to the startup ecosystem. In general, the map is a good way of identifying the hotspots of funding globally and areas with high growth or low investment infiltration.

5. Diversity Global Map of industry (Color-Coded Industry Points):

Illustrates the distribution of various industries across the globe with equal multi-industry centers in the U.S and Europe and upcoming niches in Asia and others.



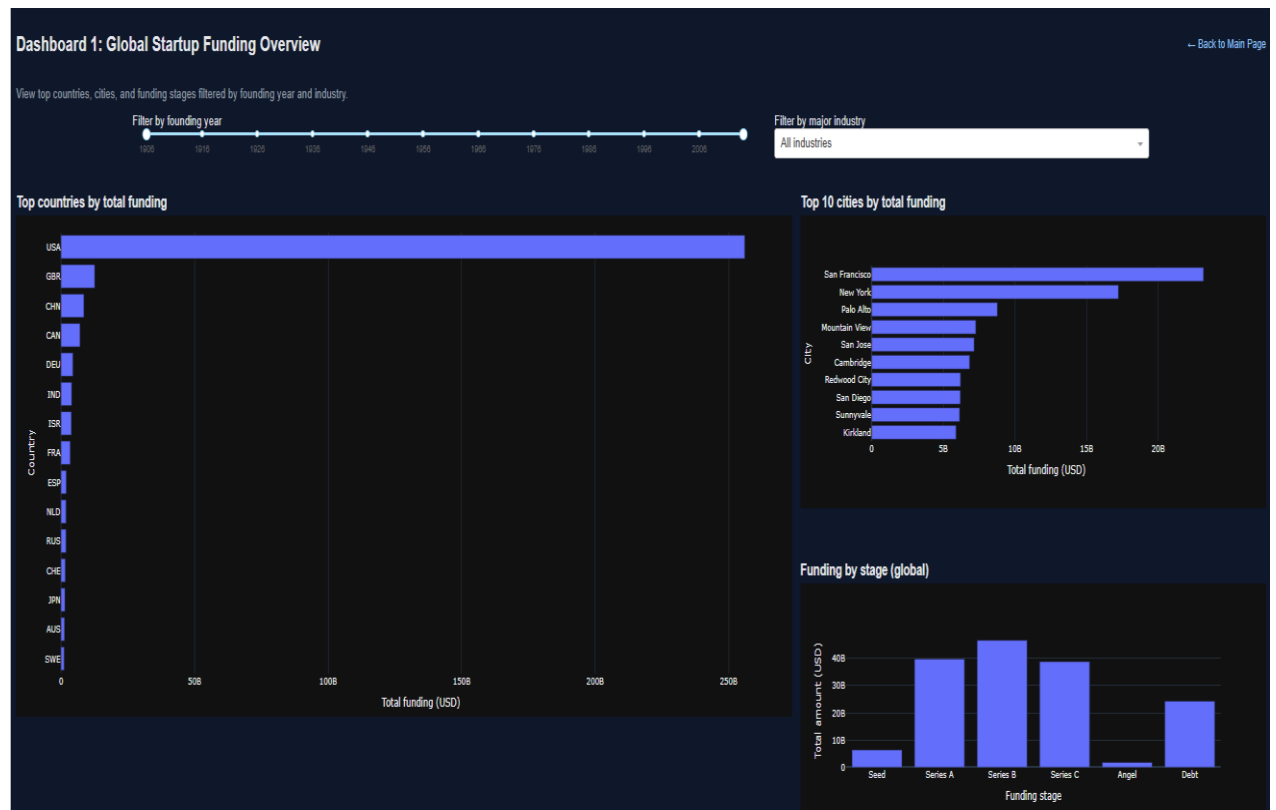
The map shows the distribution of startups around the world by major industry as separate color-coded points where the distribution of various sectors in the world can be seen. The North American and Europe are characterized by thick and diverse clusters, an indication of mature ecosystems that are well balanced in terms of technology, finance, health, consumer services and that are in the early stages of development like Cleantech and Gaming. Such cities as San Francisco, New York, London, Berlin, and Paris have a dynamic combination of industries, meaning that there is a wide spectrum of innovation activity, not concentrating on the single industry. Startups seem to be more visible in Asia with India, China, Singapore and Japan showing some activity in the Information Technology, Finance and Manufacturing. Elsewhere, smaller but increasing clusters can be observed in South America, Africa, and Oceania, pointing to the emergence of developing ecosystems in which some industries, usually FinTech, HealthTech, and E-commerce, are finding the following. In general, visualization shows the geographical distribution of the global startup activity as well as the industrial diversity as the driver of innovation, thus it is evident that although some of the regions are major hubs, there is a great diversity of sectors involved in the creation of the global entrepreneurial environment.

D. Interactive Dashboards:

This goal will unite all insights in one interactive Tableau dashboard where people can find trends in the industry, investment phases, and locations in real-time. The dashboard is a cohesive easy-to-use tool of analysis of the global state of startups using filters and dynamic visualizations.

1. Global Startup Funding Overview

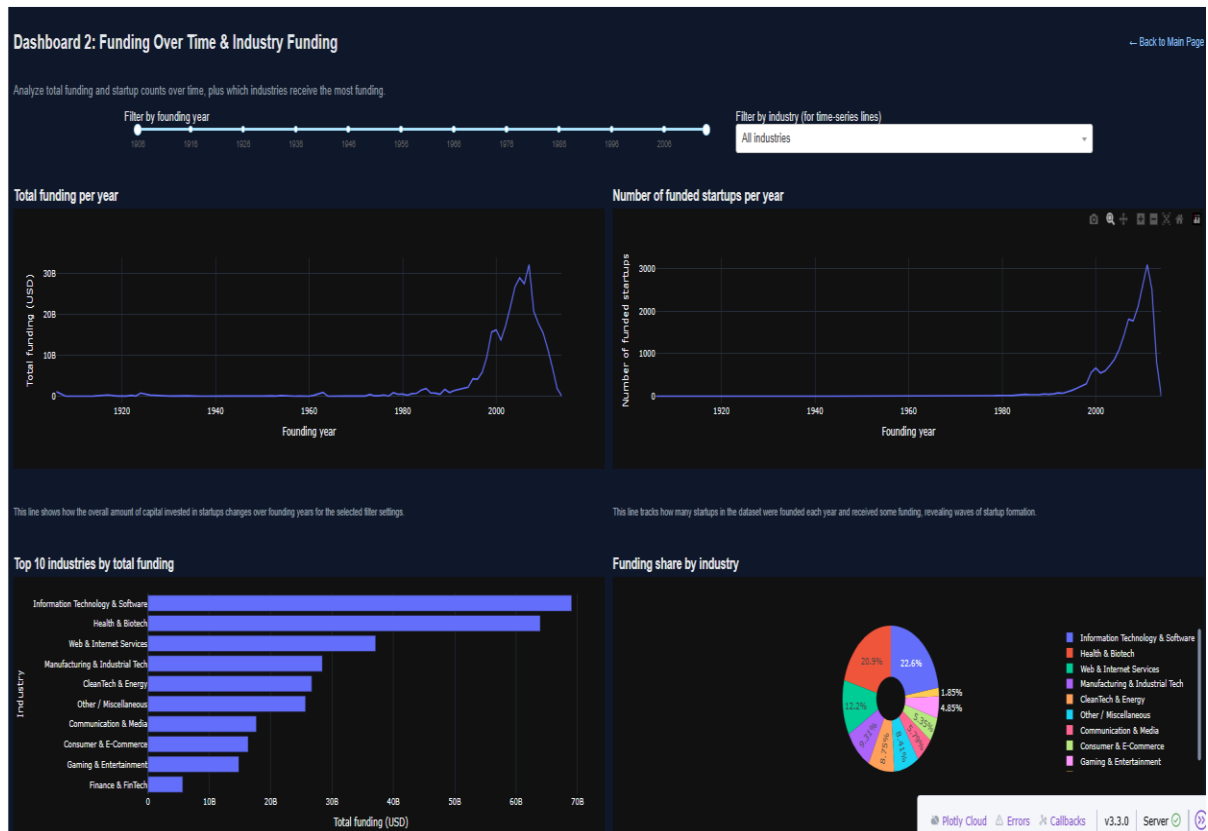
This dashboard supports two objectives because it gives geographic-level funding insights with stage-level funding summaries using bar charts with interactive filters.



This dashboard is an integrated global look at startup financing with country-level totals and top-funded cities along with the total distribution of funds across all major stages. The bar graph of the best countries as far as venture capital is concerned, shows that most of the world has its concentration on the United States, which is in excessively large proportions. The leading cities list goes even further to present certain innovation platforms including San Francisco, New York, Palo Alto, and Mountain View. The funding-by-stage bar chart in the world has summarized the amount of capital invested in the funding rounds, with the highest investment made on Series A and Series B. There are interactive filters on the founding year and industry which allow the user to dynamically interactively explore how the trends of geographic and stage-level funding change over time and industry. These elements are collectively giving the users a clear picture of venture activity distribution across the globe both at national and city levels.

2. Funding Over Time & Industry Funding:

This Dashboard gives the major trends across the timeline and the industry with interactive filters.



Dashboard 2 is devoted to the analysis of long-term trends in the activity and performance of the industry in the context of funding. The Total Funding per Year line plot depicts the history of the investment in venture capital over the decades and indicates that the growth rate was low at the beginning, followed by a significant rise in the late 1990s and 2000s, and a drop after 2012. The Number of Funded Startups per Year plot is an additional way to illustrate this since it demonstrates cycles of startup creation and their relation to variations in funding availability. The bar chart of the Top 10 Industries by Total Funding indicates those sectors which gained most cumulative investment on the head of Information Technology & Software, Health and Biotech, Web and Internet services. The Funding Share by Industry donut chart separates the contribution of each industry into the overall funding in the world. The comparison of time periods and sectors is dynamic as the user can compare both periods through the interactive filters of founding year and industry. Taken together, the dashboard displays the way the activity of startups and their focus of investors on industries and decades changed.

III. Conclusion

This project has been effective in meeting its four primary goals of converting the data of startups collected by Crunchbase into simple to understand and interactive forms. We tracked trends in the industry over long-term, visualized the activity of funding across key phases and mapped international geographic and national hot spots, as well as all of that was incorporated into a dynamic Tableau dashboard.

Industry analysis reveals that Information Technology and Software, Health and Biotech and Web and Internet services always dominate in formation of startups as well as overall investment. The trend in funding over the years shows the significant growth cycles in the 1990s, 2000s tech booms, and how different priorities of the innovation shifted over time.

Trends in funding stage show that most startups are still in their early stages, i.e. Seed and Series A, whereas Series A and B have the largest volumes of investment. This indicates great investor interest in growth-stage businesses with demonstrated traction and upscale.

Geographic mapping shows the United States is the world hub of venture activity whereby there are three major centers Silicon Valley, New York, and Boston and there are emerging ecosystems in Europe, India, China, and Southeast Asia. These hotspots show the areas of the most concentrated innovation and capital.

The built-in dashboard provides opportunities to investigate the dynamics of funding by industry, region and period. It can combine various visualizations into one analytical system, and it becomes simpler to see trends in startup activity across the world and make significant discoveries. Overall, the project offers a clear summary of venture funding development and can be helpful to entrepreneurs, investors, and scholars researching the topics of innovation and ecosystems of startups.

IV. Dataset References

[1] Justinas, “Startup Investments (Crunchbase),” Kaggle, Dataset. [Online]. Available: <https://www.kaggle.com/datasets/justinas/startup-investments>

[2] Crunchbase, “Daily CSV Export,” 2024. [Online]. Available: <https://support.crunchbase.com/hc/en-us/articles/25568433304723-Daily-CSV-Export>